Data Model and Data Dictionary: RAG-Diary

> Describes the real data structures in use: the relationships between entities, the data dictionary, indexes, search functions, and the database-level isolation policies.

> Thai version: DATA-MODEL.md

1. Document control

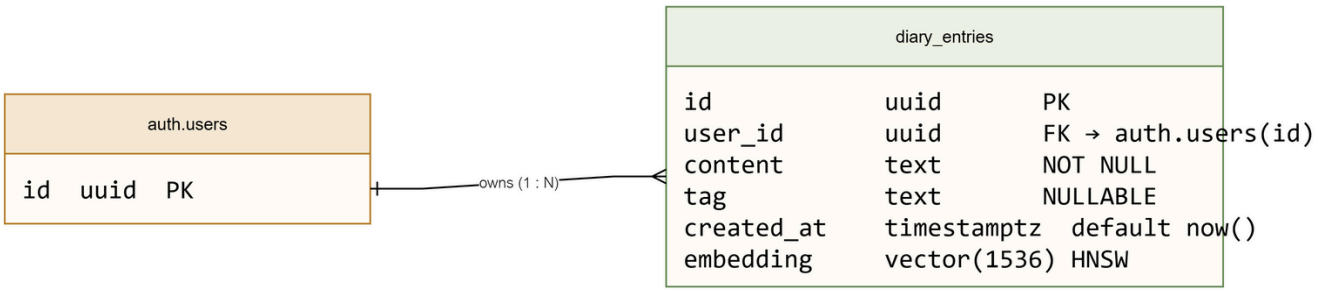
Item	Detail
Document	Data Model and Data Dictionary
Version	1.0
Prepared	15 October 2026
Author	Business Analyst / System Analyst (project owner)
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Related documents	C4-ARCHITECTURE, SRS, BUSINESS-RULES, API-CONTRACT

2. Purpose and scope

This document answers one question: how is the data stored, which tables and columns exist, and how do they relate? It shows how the data requirements became a concrete structure.

Scope: entities, relationships, the data dictionary, indexes, search functions and RLS policies. Out of scope: vector index tuning (covered by the NFRs) and API contract detail (in API-CONTRACT).

3. Entity relationship diagram



auth.users เป็นของผู้ให้บริการยืนยันตัวตน (ระบบภายนอก)
diary_entries เป็นตารางที่แอปเป็นเจ้าของ

The auth.users table belongs to the authentication provider and is managed outside the application. diary_entries is the table the application owns, so it is the focus of this document.

4. Data dictionary for diary_entries

Column	Type	Constraint	Description
id	uuid	Primary key, default gen_random_uuid()	Entry identifier
user_id	uuid	Foreign key to auth.users(id), on delete cascade	Entry owner; the basis of data isolation
content	text	Not null	Entry text
tag	text	Nullable	Entry tag (see the limit in section 8)
created_at	timestamptz	default now()	When the entry was created
embedding	vector(1536)	Nullable	Vector index of the content; may stay empty until it is built

5. Indexes

Index name	Columns and method	Purpose
diary_entries_user_id_idx	btree on user_id	Find one user's entries quickly; this is the column RLS policies filter on
diary_entries_embedding_idx	HNSW on embedding with vector_cosine_ops	Fast semantic similarity even as data grows

6. Database search functions

Function	Parameters	Behaviour
match_diary_entries	query_embedding, match_threshold, match_count	Pure semantic search: similarity is 1 minus cosine distance, filtered by the threshold
search_diary_entries	query_text, query_embedding, match_limit, keyword_weight, semantic_weight	Hybrid search: scores keyword matching and semantic similarity, then weights them as given

The hybrid function is what supports FR-09, which needs retrieval even when the exact words are forgotten, while still keeping keyword matching as a signal.

7. Row-level security policies

diary_entries has RLS enabled with four policies, one per action: select, insert, update and delete. Every policy requires auth.uid() to equal the row's user_id.

The result is that one user cannot read or change another user's data at the database level, without relying only on checks in the application. This supports FR-15 and business rules BR-01 and BR-02.

8. Observations and limits of version 1.0

Topic	Current state	Impact	Proposal
Tags	Stored as a single text column	FR-04 specifies one tag per entry, and the screen lets users create new tag names, which is enough for v1.0	To search by several tags at once, add a tags table and a join table in a later round
Empty embedding allowed	The embedding column is nullable	An entry without an embedding is not found by search	Confirm the embedding was built before treating the entry as searchable, per BR-03
Delete all	Uses DELETE /data in a single call (it used to loop per entry)	One request instead of many, even with a lot of data	Closed
Deleting a user	Uses on delete cascade	Deleting an account removes all of its entries	Matches BR-07 and the privacy requirements
Index build fails	The entry and text are saved but the embedding stays empty, so the entry will not appear in search until it is saved again	The user loses no text, but search misses it temporarily	Add a background retry queue for indexing in a later round

9. Tracing data back to requirements

Entity or column	Requirement	Business rule
user_id and the RLS policies	FR-15	BR-01, BR-02, BR-10
content	FR-04, FR-07	BR-09
tag	FR-04	BR-09
created_at	FR-06	-
embedding	FR-05, FR-09	BR-03, BR-04

10. References

Document	Relationship
C4-ARCHITECTURE	The container that stores the data
SRS	The data requirements
BUSINESS-RULES	The rules the structure must enforce
API-CONTRACT	The endpoints that read and write this data